

5.3 DTrace for Developers

The providers discussed in this section collect types of information that often are most interesting to application developers: syscall, proc, pid, sdt, vminfo. Use these providers to observe running processes as well as process creation and termination, LWP creation and termination, and signal handling. This section focuses on the pid provider.

5.3.1 The pid Provider

The pid provider instruments the entry and return from any user level function in a running process. With the pid provider you can trace any instruction in any process on the system. The name of the pid provider includes the process id of the running process that you want to examine.

5.3.2 The syscall Provider

The syscall provider reports information about system calls made. The following script displays the stack trace when a program makes a write() system call.

```
#!/usr/sbin/dtrace -s
syscall::write:entry
{
    @[ustack()]=count();
}
```

5.3.3 The sysinfo Provider

The sysinfo provider reports information about kernel statistics that are classified by the name sys. These kernel statistics provide the input for system monitoring utilities such as mpstat. The following script counts the number of times various processes get to run in the CPU. The sysinfo::pswitch event occurs when a process is switched to run on the CPU.

Enter Ctrl-C to display the results from running this script.

```
#!/usr/sbin/dtrace -s
sysinfo::pswitch
{
    @[execname] = count();
}
```

5.3.4 The proc Provider

The proc provider reports information about processes. The following script